Survival Analysis — Formula Sheet

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This sheet collects the key formulas of the course, organised by the chapters of Collett (2023). Equation numbers in parentheses, e.g. (2.4), refer to the book's own numbering; section references such as §2.2.1 point to where the formula is derived and illustrated. Formulas that the book leaves unnumbered carry only a section reference. Notation that recurs throughout is defined once below.

Notation

Symbol	Meaning
T	Survival (event) time of an individual
C	Censoring time
$X = \min(T, C)$	Observed follow-up time

Symbol	Meaning
$\delta = I(T \le C)$	Event indicator (1 = event, 0 = right-censored)
$F(t) = P(T \le t)$	Distribution function of T
$S(t) = P(T > t)$	Survivor function
$f(t)$	Probability density function of T
$h(t)$	Hazard function
$H(t)$	Cumulative hazard function
t_j	Ordered distinct event times ($j = 1, \ldots, k$)
n_j	Number at risk just before t_j
d_j	Number of events at t_j
c_j	Number censored in the interval ending at t_j
$\mathcal{R}(t)$	Risk set at t: individuals still at risk just before t
$\mathbf{x}_i$	Vector of covariates for individual i
$\boldsymbol{\beta}$	Vector of regression coefficients
$\eta_i = \boldsymbol{\beta}'\mathbf{x}_i$	Linear predictor (risk score / prognostic index)
$\psi = \exp(\beta)$	Hazard ratio

1. Survival analysis

Fundamentals, censoring, and the three key functions (§1.2, §1.4).

Distribution function (1.1):

$$F(t) = P(T \le t) = \int_0^t f(u)\,\mathrm{d}u. \qquad (1.1)$$

Survivor function (1.2):

$$S(t) = P(T > t) = 1 - F(t). \qquad (1.2)$$

Density (from the definitions in §1.4.1):

$$f(t) = \frac{\mathrm{d}F(t)}{\mathrm{d}t} = -S'(t).$$

Hazard function — the instantaneous event rate among those still at risk (1.3), equal to the density divided by the survivor function (1.4):

$$h(t) = \lim_{\delta \to 0} \frac{P(t \le T < t + \delta \mid T \ge t)}{\delta} \qquad (1.3) \qquad h(t) = \frac{f(t)}{S(t)}. \qquad (1.4)$$

Relations linking the three functions (1.5)–(1.8):

$$h(t) = -\frac{\mathrm{d}}{\mathrm{d}t}\log S(t), \qquad (1.5)$$

$$S(t) = \exp\{-H(t)\}, \qquad (1.6)$$

$$H(t) = \int_0^t h(u)\,\mathrm{d}u, \qquad (1.7)$$

$$H(t) = -\log S(t). \qquad (1.8)$$

Also $f(t) = h(t)\,S(t)$, which follows from (1.4).

Right-censoring: we observe $X = \min(T, C)$ and $\delta = I(T \le C)$. Under *independent censoring*, the censoring mechanism carries no information about T beyond the censoring time (§1.2.2).

2. Some non-parametric procedures

Kaplan–Meier, Nelson–Aalen, the life table, standard errors, and group comparisons (§2.1–2.9).

Kaplan–Meier (product-limit) estimate of the survivor function (2.4):

$$\hat{S}(t) = \prod_{t_j \le t}\left(1 - \frac{d_j}{n_j}\right) = \prod_{t_j \le t}\frac{n_j - d_j}{n_j}. \qquad (2.4)$$

Nelson–Aalen estimate of the cumulative hazard, and the survivor estimate derived from it (2.5):

$$\tilde{H}(t) = \sum_{t_j \le t}\frac{d_j}{n_j}, \qquad \tilde{S}(t) = \exp\bigl\{-\tilde{H}(t)\bigr\} = \prod_{t_j \le t}\exp(-d_j/n_j). \qquad (2.5)$$

Life-table estimate uses the effective number at risk (2.2) in the product (2.3):

$$n_j' = n_j - c_j/2 \qquad (2.2) \qquad \hat{S}(t) = \prod_{j}\left(1 - \frac{d_j}{n_j'}\right). \qquad (2.3)$$

Greenwood's formula — estimated variance (2.11) and standard error (2.12) of $\hat{S}(t)$:

$$\operatorname{var}\bigl\{\hat{S}(t)\bigr\} = \hat{S}(t)^2 \sum_{t_j \le t}\frac{d_j}{n_j(n_j - d_j)}, \qquad (2.11)$$

$$\operatorname{se}\bigl\{\hat{S}(t)\bigr\} = \hat{S}(t)\sqrt{\sum_{t_j \le t}\frac{d_j}{n_j(n_j - d_j)}}. \qquad (2.12)$$

Confidence intervals: the plain Wald interval $\hat{S} \pm 1.96\,\mathrm{se}$ can leave $[0, 1]$; a log or log-log transformation is preferred in practice (§2.2.3).

Estimated median survival time: the smallest t for which $\hat{S}(t) < 0.5$ (§2.4).

Log-rank test for two groups (§2.6.2). With $e_{1j} = n_{1j} d_j / n_j$ the expected events in group 1 under the null (2.19),

$$e_{1j} = n_{1j} d_j / n_j, \qquad (2.19)$$

$$U_L = \sum_j (d_{1j} - e_{1j}), \qquad (2.20)$$

$$v_{1j} = \frac{n_{1j} n_{2j} d_j (n_j - d_j)}{n_j^2 (n_j - 1)}, \qquad (2.21)$$

$$V_L = \sum_j v_{1j}, \qquad \frac{U_L^2}{V_L} \sim \chi^2_1. \qquad (2.23)$$

The Wilcoxon test weights each contribution by n_j (more sensitive to early differences); the Peto–Peto test weights by an estimate of $S(t)$ (preferred when censoring patterns differ) (§2.6.3–2.6.5). Stratified tests combine the U and V contributions over strata (2.25); the log-rank test for trend uses weighted sums over ordered categories (2.26)–(2.27) (§2.8–2.9).

3. The Cox regression model

The proportional hazards model, partial likelihood, and inference (§3.1–3.14).

Model for the hazard of individual i (general form, 3.3):

$$h_i(t) = h_0(t) \exp(\boldsymbol{\beta}' \mathbf{x}_i) = h_0(t) \exp(\eta_i). \qquad (3.3)$$

The baseline hazard $h_0(t)$ is left unspecified (semi-parametric). For two treatment groups (3.1) gives $h_N(t) = \psi \, h_S(t)$, i.e. a constant hazard ratio $\psi = \exp(\beta)$.

Partial likelihood (no ties); the product runs over event times with risk sets $\mathcal{R}(t_i)$ (3.5), with log-likelihood (3.6):

$$L(\boldsymbol{\beta}) = \prod_{i:\delta_i = 1} \frac{\exp(\eta_i)}{\sum_{\ell \in \mathcal{R}(t_i)} \exp(\eta_\ell)}, \qquad (3.5)$$

$$\log L(\boldsymbol{\beta}) = \sum_{i:\delta_i = 1} \left\{ \eta_i - \log \sum_{\ell \in \mathcal{R}(t_i)} \exp(\eta_\ell) \right\}. \qquad (3.6)$$

Estimation maximises $\log L$ (Newton–Raphson using the score U and information I); $\mathrm{se}(\hat\beta_j) = (I^{-1})_{jj}^{1/2}$ (§3.3.3).

Hazard ratio and its confidence interval (§3.4, §3.4.1): a $100(1-\alpha)\%$ interval for β is

$$\hat\beta \pm z_{\alpha/2} \, \mathrm{se}(\hat\beta),$$

then exponentiate to obtain the interval for ψ. The standard error of $\hat\psi$ follows from (3.13):

$$\operatorname{se}(\hat{\psi}) = \hat{\psi}\operatorname{se}(\hat{\beta}). \qquad (3.13)$$

Model comparison (§3.5): the statistic $-2\log\hat{L}$ is compared between nested models,

$$\Delta(-2\log\hat{L}) \sim \chi^2_{p-q},$$

where $p-q$ is the difference in the number of parameters. Akaike's information criterion (3.14):

$$\mathrm{AIC} = -2\log\hat{L} + 2q. \qquad (3.14)$$

Estimated survivor function for a given covariate pattern (§3.10, 3.26):

$$\hat{S}_i(t) = \left\{\hat{S}_0(t)\right\}^{\exp(\hat{\eta}_i)}. \qquad (3.26)$$

With no covariates this reduces to the Kaplan–Meier estimate (§3.10.1).

Further tools: the lasso adds the penalty $\lambda\sum_j|\beta_j|$ to the partial log-likelihood (3.16); fractional polynomials model non-linearity (§3.8); the concordance c-index and R^2-type measures assess predictive ability (§3.12); time-dependent ROC curves assess discrimination over time (§3.13). The Cox score test is identical to the log-rank test when there is a single binary covariate (§3.14).

4. Model checking in the Cox regression model

Residuals and the proportional hazards assumption (§4.1–4.5).

Cox–Snell residuals (4.1):

$$r_{Ci} = \hat{H}_0(t_i)\exp(\hat{\eta}_i) = -\log\hat{S}_i(t_i). \qquad (4.1)$$

Martingale residuals (4.6):

$$r_{Mi} = \delta_i - r_{Ci}. \qquad (4.6)$$

Deviance residuals (4.7):

$$r_{Di} = \operatorname{sign}(r_{Mi})\sqrt{-2[r_{Mi} + \delta_i\log(\delta_i - r_{Mi})]}. \qquad (4.7)$$

Schoenfeld residuals for covariate j at each event time (4.8):

$$r_{Sji} = \delta_i\{x_{ji} - \hat{a}_{ji}\}. \qquad (4.8)$$

Score residuals (4.13) are the building blocks of influence diagnostics (§4.3). The proportional hazards assumption is tested from the Schoenfeld residuals: under PH, $E(r_{Sji}) \approx \beta_j(t_i) - \hat{\beta}_j$ (4.18), so a systematic trend against time leads to the test statistic (4.19)–(4.20) (§4.4.2–4.4.3). Adding a time-dependent variable (4.22) is the regression-based alternative (§4.4.4).

5. Parametric regression models

Fully parametric hazard and accelerated failure time models (§5.1–5.17).

Common distributions (hazard $h(t)$, survivor $S(t)$):

Distribution	$h(t)$	$S(t)$
Exponential	λ (5.2)	$\exp(-\lambda t)$ (5.2)
Weibull	$\lambda\gamma t^{\gamma-1}$ (5.4)	$\exp(-\lambda t^{\gamma})$ (5.5)
Log-logistic	$\frac{\lambda\gamma(\lambda t)^{\gamma-1}}{1+(\lambda t)^{\gamma}}$ (5.8)	$\frac{1}{1+(\lambda t)^{\gamma}}$ (5.9)
Lognormal	— (§5.1.4)	$1-\Phi\left(\frac{\log t-\mu}{\sigma}\right)$ (5.10)
Gompertz	$\lambda\exp(\theta t)$ (§5.1.5)	$\exp\left\{-\frac{\lambda}{\theta}(e^{\theta t}-1)\right\}$ (§5.1.5)

Here $\lambda > 0$ is a scale and $\gamma > 0$ a shape parameter ($\gamma = 1$ recovers the exponential); gamma and inverse Gaussian are further options (§5.1.6–5.1.8). A Weibull diagnostic plot uses (5.12):

$$\log\{-\log S(t)\} = \log\lambda + \gamma\log t. \qquad (5.12)$$

Weibull proportional hazards model (5.42)–(5.43):

$$h_i(t) = \gamma t^{\gamma-1}\exp(\boldsymbol{\beta}'\mathbf{x}_i), \qquad (5.42)$$

$$S_i(t) = \exp\{-\exp(\boldsymbol{\beta}'\mathbf{x}_i)\,\lambda t^{\gamma}\}. \qquad (5.43)$$

Accelerated failure time (AFT) models (§5.10–5.12). General hazard form (5.60) and log-linear form (5.61):

$$h_i(t) = e^{-\eta_i}\,h_0(t/e^{\eta_i}), \qquad (5.60)$$

$$\log T_i = \mu + \boldsymbol{\beta}'\mathbf{x}_i + \sigma W_i, \qquad (5.61)$$

where W has a standard distribution. The Weibull is the only distribution that is simultaneously PH and AFT.

Proportional odds model (5.76):

$$\frac{S_i(t)}{1-S_i(t)} = e^{\eta_i}\frac{S_0(t)}{1-S_0(t)}. \qquad (5.76)$$

Cure-rate model (5.79):

$$S(t) = (1-\pi)S_n(t) + \pi. \qquad (5.79)$$

6. Flexible parametric models

Piecewise and spline-based models (§6.1–6.5).

Piecewise exponential model: constant hazard within each interval (6.1),

$$h_0(t) = \lambda_j \quad \text{on interval} \quad j. \qquad (6.1)$$

Restricted cubic spline (RCS) in t (6.7)–(6.8):

$$s(t) = \gamma_0 + \gamma_1 t + \sum_j \gamma_{j+1} \nu_j(t), \qquad (6.7)$$

$$\nu_j(t) = (t - k_j)_+^3 - \lambda_j (t - k_{\min})_+^3 - (1 - \lambda_j)(t - k_{\max})_+^3. \qquad (6.8)$$

Royston–Parmar model: the log cumulative hazard is an RCS in $y = \log t$ (6.13):

$$\log H_i(t) = \eta_i + \gamma_0 + \gamma_1 y + \sum_j \gamma_{j+1} \nu_j(y). \qquad (6.13)$$

7. Model checking in parametric models

Residuals and diagnostics for parametric models (§7.1–7.5).

Standardised residuals (7.1):

$$r_{Si} = \frac{\log t_i - \hat{\mu} - \hat{\boldsymbol{\beta}}' \mathbf{x}_i}{\hat{\sigma}}. \qquad (7.1)$$

Cox–Snell (7.2), martingale (7.4) and deviance (7.5) residuals carry over, now based on the fitted parametric $S(t)$ and $H(t)$:

$$r_{Ci} = \hat{H}_i(t_i) = -\log \hat{S}_i(t_i), \qquad (7.2)$$

$$r_{Mi} = \delta_i - r_{Ci}, \qquad (7.4)$$

$$r_{Di} = \operatorname{sign}(r_{Mi}) \sqrt{-2[r_{Mi} + \delta_i \log(\delta_i - r_{Mi})]}. \qquad (7.5)$$

Fitted and observed survivor functions are compared visually (§7.3), and influential observations are identified from the score residuals (§7.4).

8. Time-dependent variables

Covariates and coefficients that change over time (§8.1–8.7).

Time-dependent covariate $x(t)$ (8.1), with survivor function (8.4):

$$h_i(t) = h_0(t) \exp\{\boldsymbol{\beta}' \mathbf{x}_i(t)\}, \qquad (8.1)$$

$$S_i(t) = \exp\left[-\int_0^t \exp\{\boldsymbol{\beta}' \mathbf{x}_i(u)\}\, h_0(u)\, \mathrm{d}u \right]. \qquad (8.4)$$

Time-varying coefficient $\beta(t)$ allows a covariate's effect to change over time (§8.1.1). Joint models couple a longitudinal submodel for repeated measurements with a survival submodel (8.10):

$$h_i(t) = \exp\{\boldsymbol{\beta}' \mathbf{x}_i + \alpha\, m_i(t)\}\, h_0(t). \qquad (8.10)$$

9. Interval-censored survival data

When the event time is known only up to an interval (§9.1–9.4).

The non-parametric maximum likelihood estimate (Turnbull) of $S(t)$ puts mass $\hat{\theta}_j$ on the Turnbull intervals $(a_j, b_j]$ and is (9.2):

$$\hat{S}(t) = \prod_{t_j \le t} \hat{\theta}_j. \tag{9.2}$$

The likelihood for interval-censored data is (9.3):

$$L = \prod_{i} \{ S(l_i) - S(u_i) \}. \tag{9.3}$$

Semi-parametric (piecewise exponential) and parametric alternatives are described in §9.3–9.4.

10. Frailty models

Unobserved heterogeneity and clustered data (§10.1–10.8).

Individual frailty Z_i scales the hazard (10.2):

$$h_i(t \mid Z_i) = Z_i \, h_0(t) \exp(\boldsymbol{\beta}' \mathbf{x}_i). \tag{10.2}$$

With gamma frailty $Z_i \sim \text{Gamma}(1/\theta, 1/\theta)$ (mean 1, variance θ), the observable survivor function is (10.12):

$$S_i^*(t) = \left\{ 1 + \theta^{-1} e^{\boldsymbol{\beta}' \mathbf{x}_i} H_0(t) \right\}^{-\theta}. \tag{10.12}$$

The shared frailty model assigns a common Z_i to all members of a cluster (10.31):

$$h_{ij}(t \mid Z_i) = Z_i \, h_0(t) \exp(\boldsymbol{\beta}' \mathbf{x}_{ij}). \tag{10.31}$$

11. Non-proportional hazards and institutional comparisons

Stratification, restricted mean survival, and random effects (§11.1–11.4).

Stratified proportional hazards model — separate baseline hazard per stratum, common coefficients (§11.2):

$$h_{sk}(t) = h_{0k}(t) \exp(\boldsymbol{\beta}' \mathbf{x}).$$

Restricted mean survival time over $[0, t_0]$ (§11.3):

$$\mu(t_0) = E\{\min(T, t_0)\} = \int_0^{t_0} S(u) \, \mathrm{d}u.$$

The pseudo-value for individual i is (11.4):

$$z_i = n \, \hat{\mu}(t_0) - (n-1) \, \hat{\mu}_{(-i)}(t_0). \tag{11.4}$$

Institutional comparisons use random-effect (frailty) or Poisson formulations, comparing observed to expected failures via (11.8) (§11.4).

12. Competing risks

Cause-specific hazards and the cumulative incidence function (§12.1–12.6).

Cause-specific hazard for cause j (12.1):

$$\lambda_j(t) = \lim_{\delta \to 0} \frac{P(t \le T < t+\delta, K = j \mid T \ge t)}{\delta}. \qquad (12.1)$$

Cumulative incidence function — the probability of an event *of cause* j by time t (12.4):

$$F_j(t) = \int_0^t \lambda_j(u)\, S(u)\, \mathrm{d}u, \qquad (12.4)$$

estimated by (12.5):

$$\hat{F}_j(t) = \sum_{t_i \le t} \frac{\delta_{ij}}{n_i}\, \hat{S}(t_{i-1}). \qquad (12.5)$$

Fine–Gray subdistribution hazard (12.10) and its proportional model (12.11):

$$\gamma_j(t) = -\frac{\mathrm{d}}{\mathrm{d}t} \log\bigl\{1 - F_j(t)\bigr\}, \qquad (12.10)$$

$$\lambda_{ij}(t) = \exp(\boldsymbol{\beta}_j' \mathbf{x}_i)\, \lambda_{0j}(t). \qquad (12.11)$$

Note that $1 - \sum_j F_j(t) = S(t)$; the cumulative incidence of one cause depends on the hazards of all causes.

13. Multiple events and event history modelling

Counting processes and recurrent events (§13.1–13.5).

A counting process $N_i(t)$ counts events up to t; its intensity is (13.1), which for survival data reduces to the at-risk indicator times the hazard (13.4):

$$\lambda_i(t)\, \mathrm{d}t = Y_i(t)\, h_i(t)\, \mathrm{d}t. \qquad (13.4)$$

The martingale is $M_i(t) = N_i(t) - \Lambda_i(t)$ (13.3). Models for recurrent events: Anderson–Gill (13.5) and Prentice–Williams–Peterson (13.6):

$$h_i(t) = Y_i(t) \exp\{\boldsymbol{\beta}' \mathbf{x}_i(t)\}\, h_0(t), \qquad (13.5)$$

$$h_{ij}(t) = Y_{ij}(t) \exp\bigl\{\boldsymbol{\beta}_j' \mathbf{x}_i(t)\bigr\}\, h_{0j}(t). \qquad (13.6)$$

Wei–Lin–Weissfeld uses marginal models per event (§13.3). Robust (sandwich) standard errors accommodate within-subject correlation (§13.1.4).

14. Dependent censoring

When censoring is informative (§14.1–14.3).

Inverse probability of censoring weighting (IPCW) reweights individuals by the inverse of the estimated probability of remaining uncensored, obtained from a censoring model (14.3), entering the weighted partial likelihood (14.4):

$$\hat{S}_{ci}(t) = \left\{\hat{S}_{c0}(t)\right\}^{\exp(\hat{\beta}_c' \mathbf{x}_i)}. \tag{14.3}$$

15. Sample size requirements for a survival study

Number of events and number of patients (§15.1–15.3).

Required number of events (deaths) to detect a hazard ratio ψ with two-sided level α and power $1-\beta$, allocation proportions π_1, π_2 (15.1); equal allocation gives (15.2):

$$d = \frac{(z_{\alpha/2} + z_{\beta})^2}{\pi_1 \pi_2 \, (\log \psi)^2}, \tag{15.1}$$

$$d = \frac{4(z_{\alpha/2} + z_{\beta})^2}{(\log \psi)^2}. \tag{15.2}$$

The required number of patients follows by inflating for the probability of an event by the end of follow-up (15.10):

$$n = \frac{d}{P(\text{event})}, \tag{15.10}$$

where $P(\text{event})$ is obtained from the survivor function, e.g. via Simpson's rule (15.11) (§15.3).

16. Bayesian survival analysis

Priors, posteriors, and model comparison (§16.1–16.10).

Bayes' theorem for the model parameters $\boldsymbol{\theta}$ (16.4):

$$p(\boldsymbol{\theta} \mid \text{data}) = \frac{L(\boldsymbol{\theta}) \, p(\boldsymbol{\theta})}{\int L(\boldsymbol{\theta}) \, p(\boldsymbol{\theta}) \, \mathrm{d}\boldsymbol{\theta}}. \tag{16.4}$$

The likelihood for survival data is (16.5):

$$L(\boldsymbol{\theta} \mid t) = \prod_i \{h_i(t_i)\}^{\delta_i} \, S_i(t_i). \tag{16.5}$$

Model comparison uses the deviance information criterion (16.15) and the widely applicable information criterion (16.16):

$$\mathrm{DIC} = D(\bar{\boldsymbol{\theta}}) + 2p_D, \tag{16.15}$$

$$\mathrm{WAIC} = -2 \sum_i \log p(t_i \mid t) + 2p_W. \tag{16.16}$$

17. Survival analysis with R

Mapping the methods to the `survival` package (§17.1–17.16).

Task	Function
Create a survival object	`Surv(time, event)`
Kaplan–Meier / Nelson–Aalen curves	`survfit()`
Log-rank and related tests	`survdiff()`
Cox regression	`coxph()`
PH diagnostics and residuals	`cox.zph()`, `residuals(..., type = "...")`
Parametric models (PH and AFT)	`survreg()`
Flexible (Royston–Parmar) models	`flexsurv::flexsurvspline()`
Competing risks (Fine–Gray)	`cmprsk::crr()` / `tidycmprsk`
Kaplan–Meier plots	`ggsurvfit::ggsurvfit()`

The parametric log-linear form (17.1) and the frailty model (17.2) restate (5.61) and (10.2) in the notation used by `survreg` and `coxph`:

$$\log T_i = \mu + \boldsymbol{\alpha}'\mathbf{x}_i + \sigma\varepsilon_i, \qquad (17.1)$$

$$h_i(t) = z_i \exp(\boldsymbol{\beta}'\mathbf{x}_i)\, h_0(t). \qquad (17.2)$$

Appendices

Appendix A summarises maximum likelihood estimation for one and several parameters (score, information, standard errors) — the machinery behind the estimators above. Appendix B lists five additional data sets (chronic active hepatitis, bladder cancer, black ducks, bone marrow transplantation, chronic granulomatous disease) that can be used for practice analyses.